

Full paper

Conductive silver nanowires-fenced carbon cloth fibers-supported layered double hydroxide nanosheets as a flexible and binder-free electrode for high-performance asymmetric supercapacitors



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ABSTRACT

Silver nanowires (Ag NWs) have attracted particular interest in the development of various electronic and energy storage devices due to their one-dimensional structure, good conductivity, fast charge transportation and direct contact to the current collector. Herein, we have successfully deposited the binder-free nickel-cobalt layered double hydroxide nanosheets on Ag NWs-fenced carbon cloth (NC LDH NSs@Ag@CC) by a facile electrochemical deposition method with a chronoamperometry voltage of -1.0 V for 120 s. The electrically conductive and superhydrophilic nature of the hybrid nanocomposite electrode led to relatively high areal capacitance (1133.3 mF cm $^{-2}$ at 1 mA cm $^{-2}$) and good cycling stability (80.47% after 2000 cycles) compared to the electrode prepared without Ag NWs. Using such hierarchical NC LDH NSs@Ag@CC as a positive electrode, we further fabricated a flexible asymmetric supercapacitor (ASC) with activated carbon coated CC as a negative electrode. The as-assembled ASC exhibited maximum operational potential window of 1.6 V, high areal capacitance of 230.2 mF cm $^{-2}$ and excellent cycling stability of 88.1% with remarkable energy densities at all the charge-discharge conditions (78.8 μWh cm $^{-2}$ at the power density of 785 μW cm $^{-2}$ and 40 μWh cm $^{-2}$ at the high power density of 12.1 mW cm $^{-2}$, respectively).

1. Introduction

Recently, supercapacitors (SCs) have attracted appreciable attention as a new class of energy storage device owing to their inherent advantages of high power density, fast charge-discharge speed, long cycling lifetime and low cost [1–3]. Accordingly, SCs are utilized in various fields, including hybrid vehicles, back-up power systems, portable electronic devices, etc [4–6]. Although the SCs have many benefits over the batteries and fuel cells, they are still suffering from poor energy density problem [7–9]. Consequently, great attempts have been made for achieving the high energy density in SCs without affecting the power density. From the equation of $E = \frac{1}{2} CV^2$, it is evident that the energy density can be enhanced by maximizing the capacitance and working potential window of the device [10–12]. The construction of asymmetrical SCs (ASCs) is one of the promising approaches for achieving the superior energy density characteristics [13]. This ASC can be assembled by combining the two kinds of materials, namely, the battery-type electroactive material as an energy source and the electric-double layer capacitive electroactive material as a power source, respectively [14–16]. Thus, these two electroactive materials in the

device could utilize their entire well-separated potential windows, resulting in the higher operating voltage and delivering the remarkable capacitance and significantly enhanced energy density [17].

Up to now, several transition metal oxides/hydroxides such as Co_3O_4 , NiWO_4 , $\text{Co}(\text{OH})_2$, $\text{Ni}(\text{OH})_2$, NiMoO_4 , etc. have been widely explored as a battery-type or redox active material in ASCs, by virtue of their high charge storing capability and enhanced electrochemical activity [18–21]. Recently, several transition metallic layered double hydroxides (LDHs), usually represented by the formula $[\text{M}^{2+}_{1-x}\text{M}^{3+}_x(\text{OH})_2][\text{A}^{n-}_{x/n}\cdot m\text{H}_2\text{O}]$, where M^{2+} and M^{3+} are divalent and trivalent metal cations, respectively and A^{n-} could be mostly organic or inorganic anions (OH^- , CO_3^{2-} , NO_3^- , etc.), have been extensively investigated due to their high anion exchanging property and capability of intercalating anions [22–24]. In general, the metallic LDHs are composed of nanometer-thick layered structure, in which excessive positive charges corresponding to the trivalent or twice the tetravalent metal ions are compensated by the anions [25]. Such intrinsic properties of high surface area and tunable metal compositions of LDHs without varying their structure make them extensively used in the various fields, such as catalysis, anion exchanging, batteries,

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energy storage applications, etc [23,26]. Moreover, these materials comprising of transition metals have been extensively utilized as battery-type materials for SCs due to their enrich slabs, high electrochemical conductivity and versatile oxidation states within the transition metal atoms [24,27]. The anion exchange capacity in metallic LDH materials is greatly enhanced by the intercalating anions and water molecules into a large interlayer spacing of unique two-dimensional (2D) lamellar structures [28]. Due to the faradaic redox reactions with aqueous electrolytes, large electric charges may be collected on the surface of 2D thin layers and hence a high increment in the electrochemical performance can be observed [29]. Up to now, several combinations of LDH have been studied for supercapacitor applications, such as nickel-cobalt (NC), cobalt-aluminum, nickel-manganese, magnesium-aluminum LDHs, etc [24,30–32]. Among these LDHs, NC LDHs have received great interest because of their layered structures with large interlayer arrangement, high redox activity, multiple oxidation states and eco-friendly nature [33,34]. Particularly, NC LDHs with nanosheets/nanoflakes-like structures with high surface area and interconnected arrangement provide the effective paths for electrolyte diffusion, leading to the rapid redox reactions for enhanced electrochemical properties [35,36]. So far, these NC LDH nanosheets (NSs) have been prepared using several growth methods, in which the electrochemical deposition has attracted much attention owing to the low-temperature growth, short reaction time and safe apparatus [37]. Generally, conductive nickel foam, nano porous gold, metal foil, etc. based electrodes can be used as substrates for the binder-free growth of electroactive materials for SCs, but their rigidity and complicated manufacturing process imposes a limit on the development of flexible devices [38–41]. Therefore, carbon cloth (CC), graphene coated textiles and copper coated polyester fabrics based electrodes are gaining prominence as a current collector for the development of flexible/wearable SCs [19,42]. Among these substrates, the CC ensued the following advantages; high flexibility and excellent chemical/thermal stability compared to the other flexible fabrics [43]. Although CC has desirable flexible features, but it still suffers from relatively hydrophobicity and low conductive property compared to the conventional metal electrodes, which only offers a limited specific surface area platform and poor porosity for higher electroactive material loading [44,45]. To make use of CC as a promising substrate for high-performance energy storage devices, growth of conductive materials on CC could be expected to enhance the conductivity and hydrophilic property.

As is well known, silver (Ag) has the highest electrical conductivity (6.3×10^7 S/m) among the miscellaneous metals, which facilitates the direct electronic conduction [46,47]. Accordingly, the Ag based nanostructures (particularly, Ag nanowires (NWs)) were investigated as the effective collector for improving the electrical conductivity as well as the electrochemical performance in SCs [48,49]. For example, Yu et al. reported that an electrode with sandwich-like Ag NW/PEDOT: PSS-nanopillar/MnO₂ composite exhibited much higher specific capacitance owing to larger electrochemical conductivity of the nanocomposite [50]. Yuksel et al. integrated the coaxial Ag NW@Ni(OH)₂ core-shell structure, in which the conductive Ag NW core helps to increase the conductivity to obtain the larger electrochemical capacitance [51]. Furthermore, Wu et al. synthesized Ag NW coupled Ni-Al LDH NSs composite material and their pseudocapacitance was comparatively higher than that of bare Ni-Al LDH NSs [52]. Besides, Qiao et al. also prepared the hierarchical network of MnO₂@Ag NWs composite with much larger specific capacitance compared to the pristine MnO₂ without Ag NWs [53]. The high conductive nature of Ag NW promotes the rapid reaction kinetics during the energy storage measurements.

Considering their thin diameter, high aspect ratio and superior conductivity, in this work, we synthesized Ag NWs on CC (Ag@CC) using a simple dip-coating approach, eventually improving its hydrophilic nature and conductivity. The 3D network of Ag NWs on CC (Ag@CC) could be also expected to increase the mass loading of electroactive

material. Moreover, the high conductive nature of Ag NWs provides the effective pathways for electron transportation during the energy storage process [51,52]. Using a facile electrochemical deposition (ECD) technique, we further deposited NC LDH NSs on Ag@CC substrate (i.e. NC LDH NSs@Ag@CC) as a battery-type electrode. The electrochemical properties of the binder-free NC LDH NSs@Ag@CC showed high areal capacitance with good rate capability and long cycling lifetime in 1 M KOH electrolyte solution. Furthermore, the ASC was also fabricated for practical applications using NC LDH NSs@Ag@CC as a battery-type electrode and the activated carbon (AC) as an electric-double layer capacitive electrode, respectively. The energy storage properties of the ASCs device were characterized in detail.

2. Experimental section

2.1. Chemicals

Nickel nitrate hexahydrate (Ni(NO₃)₂·6H₂O) and nitric acid (HNO₃) were received from Daejung Chemicals Ltd., South Korea. Cobalt nitrate hexahydrate (Co(NO₃)₂·6H₂O) and potassium hydroxide (KOH) were obtained from Sigma-Aldrich Co., South Korea. Ag NWs suspension in isopropanol was purchased from PlasmaChem GmbH Co. The CC was purchased from Nara Cell-Tech Corp., South Korea. All the chemicals were of analytical grade and used directly without any further purification after purchasing.

2.2. Synthesis of NC LDH NSs@Ag@CC

The hierarchical NC LDH NSs were uniformly deposited on Ag@CC substrate by a simple and one-step ECD method at room temperature (RT). Initially, the bare CC was sonicated in concentrated HNO₃ solution for 10 min and dried in an oven for one hour after cleaning thoroughly with de-ionized (DI) water for several times. Then, the well-cleaned CC was cut into the rectangular pieces with sizes of 1×2.5 cm², dipped directly into Ag NWs solution two times and thermally annealed at 200 °C under argon atmosphere to improve the adhesion and conductivity. The as-prepared Ag@CC substrate was attached to the polyethylene terephthalate (PET) film for restricting the deposition of NC LDH NSs on the back side of the substrate. After that, the ECD process was carried out in a conventional beaker-type three-electrode cell, which consists of Ag@CC as a working, saturated calomel electrode (SCE) as a reference, and platinum (Pt) wire as a counter electrode, respectively. Here, the total area of Ag@CC (1×2.5 cm²) is divided into two parts using a plastic tape, in such a way that the 1×2 cm² active area is utilized for the deposition of NC LDH and the remaining part (1×0.5 cm²) is assigned for electrode contact, respectively. Meanwhile, the growth solution was prepared by mixing appropriate amounts of Ni(NO₃)₂·6H₂O and Co(NO₃)₂·6H₂O (0.067 and 0.033 M, respectively) in a glass beaker containing 40 mL of DI water under constant magnetic stirring at RT. Then, the ECD process was performed after carefully immersing the three-electrode system into growth solution with an applied cathodic voltage of -1.0 V for 120 s (chronoamperometry technique) using an IviumStat electrochemical workstation (The Netherlands). After ECD process, the NC LDH NSs@Ag@CC was removed from experimental set-up and washed gently with DI water to remove the partly adhered NC LDH NSs on substrate. Then, the sample was dried under the flow of pure nitrogen gas and kept in a vacuum oven at 80 °C for 2 h. The mass loading of NC LDH NSs on Ag@CC was obtained to be ~ 0.7 mg/cm². For comparison, the NC LDH NSs were also deposited on bare CC using the same ECD process under similar conditions and it was designated as NC LDH NSs@CC (~ 0.3 mg/cm²).

2.3. Characterization

The surface morphologies and nanostructures of the prepared

samples were characterized by using a field-emission scanning electron microscope (FE-SEM, Carl Zeiss, LEO SUPRA 55) and a high-resolution transmission electron microscope (HR-TEM) equipped with an energy dispersive X-ray spectroscopy (EDX). The crystal structure of the synthesized material was analyzed by an X-ray diffraction (XRD) pattern. The information regarding to vibrational modes of the sample was evaluated by a high-resolution Raman (HR-Raman) spectroscopy. The surface chemical distribution and oxidation states of the elements in the prepared sample were examined by an X-ray photoelectron spectroscopy (XPS, Thermo Electron MultiLab2000) with an Al K α radiation. The nitrogen (N₂) adsorption-desorption isotherms and pore size were measured by using a BELSORP-max (MP) Brunauer-Emmett-Teller (BET) surface analyzer. A contact angle measurement system (Phoenix-300, SEO Co., Ltd.) was utilized for the measurement of wetting behavior of electrodes.

2.4. Electrochemical measurements

The electrochemical properties of the synthesized samples were measured with an IVIUMSTAT electrochemical workstation. Cyclic voltammetry (CV), galvanic charge-discharge (GCD) and electrochemical impedance spectroscopy (EIS) tests were performed in a three-electrode system with 1 M KOH aqueous electrolyte. Here, the prepared samples were used as a working electrode, the Ag/AgCl and Pt wire were employed as the reference and counter electrodes, respectively. The ASC was assembled with the NC LDH NSs@Ag@CC composite material as a positive electrode, AC coated CC (see [Supplementary Information](#)) as a negative electrode and Whatman filter paper soaked in 1 M KOH electrolyte as a separator. The formulae for calculating the areal capacitance (C_{ac} , F cm⁻²), energy density (E , Wh cm⁻²) and power density (P , W cm⁻²) of the device are given below [12,54]:

$$C_{ac} = \frac{I}{a} \times \frac{\Delta t}{\Delta V} \quad (1)$$

$$E = \frac{C_{ac}}{2} \times (\Delta V)^2 \quad (2)$$

$$P = \frac{E}{\Delta t} \quad (3)$$

where I is the discharge current (A), Δt is the discharge time (s), ΔV is the potential drop (V) and a is the active area of the electrode (cm²).

3. Results and discussion

Fig. 1 shows the schematic illustration for the preparation process of NC LDH NSs@Ag@CC. The highly flexible features with intertwined conductive fibrous framework of CC were utilized as a substrate (**Fig. 1(a)**). Then, the Ag NWs were uniformly decorated on CC fibers by a simple dip-coating technique, resulting in a 3D Ag NW network on CC, as depicted in **Fig. 1(b)**. The as-fenced Ag NWs eventually improve the conductivity as well as the hydrophilic nature of CC substrate. Afterwards, the hierarchical NC LDH NSs were electrochemically deposited on Ag@CC by using a simple three electrode system at RT. The formation of hierarchical NC LDH NSs during the ECD process in the bimetallic (Ni and Co) growth solution is described as follows. When the chronoamperometry voltage of -1.0 V is applied to the working electrode, the current passed through the growth solution reduces the nitrate (NO₃⁻) ions on the cathodic exterior surface, ensuing the generation of hydroxyl (OH⁻) ions. As a result of the liberated OH⁻ ions, the local pH value on the surface of Ag@CC is raised and hence the reaction is further prolonged. Ultimately, the bimetallic ions (Ni²⁺ and Co²⁺) in the precursor rapidly react with the OH⁻ ions at the vicinity of substrate, which intends to be formed as NC LDHs on the Ag@CC, as shown in **Fig. 1(c)**. The electrochemical and

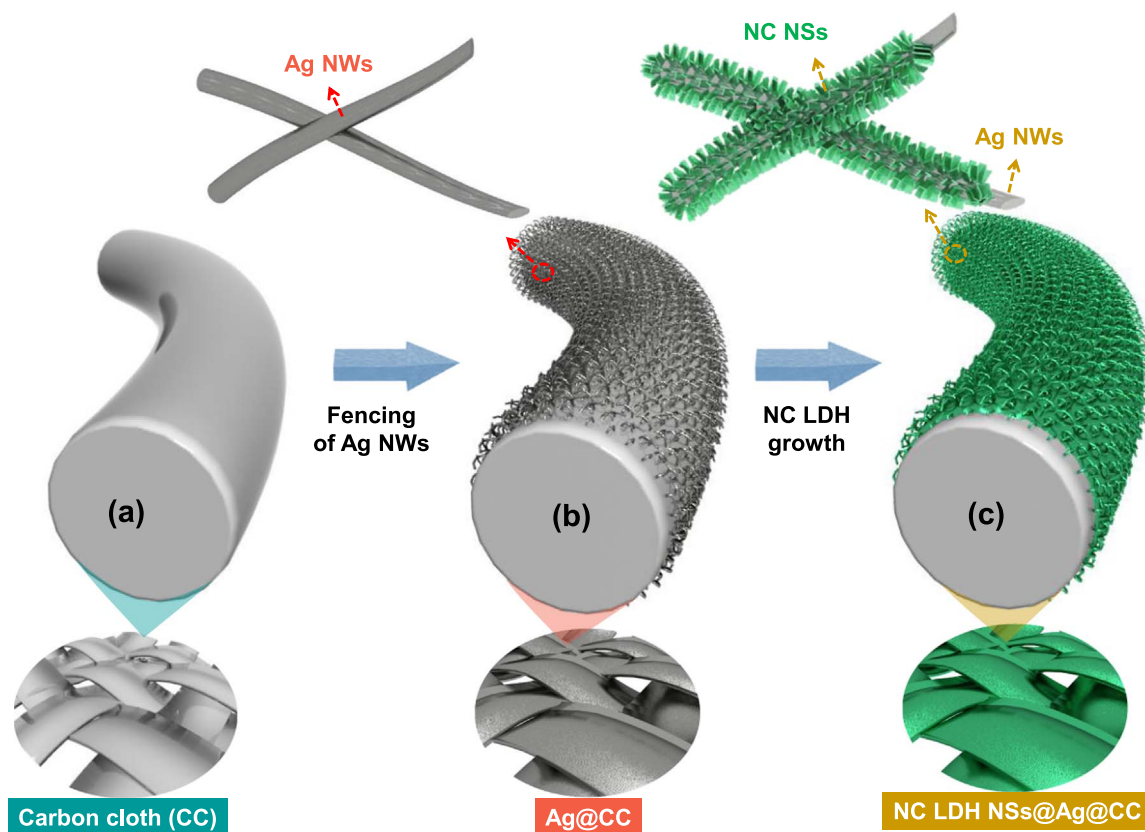


Fig. 1. (a–c) Schematic illustration showing the preparation process flow of NC LDH NSs@Ag@CC by a single-step ECD process.

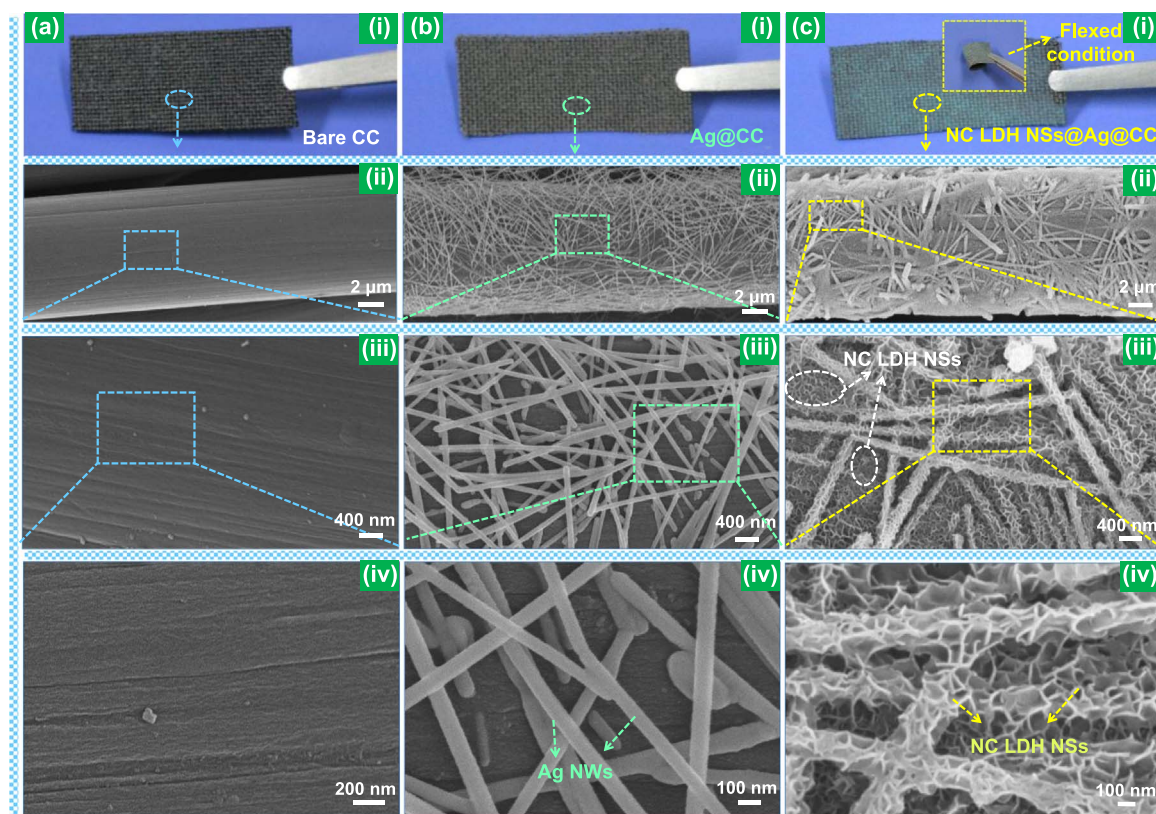


Fig. 2. Photographic and FE-SEM images of the (a) bare CC, (b) Ag@CC and (c) NC LDH NSs@Ag@CC samples.

subsequent precipitation reactions during the synthesis of NC LDH NSs are elucidated as follows [30,37,55]:



Fig. 2(a–c) displays the photographic and FE-SEM images of the bare CC, Ag@CC and NC LDH NSs@Ag@CC, respectively. As shown in Fig. 2(a) (i), the color of the bare CC looks black and it weaves with several conductive fibers in a disorder manner. The low-magnification FE-SEM image of Fig. 2(a) (ii) confirms that the CC has conductive fibers with a diameter of $\sim 9.5 \mu\text{m}$. With the increased magnification, it is observed that the CC fiber has a smooth surface (Fig. 2(b) (iii–iv)), which leads to the poor adhesion and low mass loading of the material on CC. After coating with Ag NWs, the color of the CC was slightly changed, as shown in the photographic image of Fig. 2(b) (i). From the low-magnification FE-SEM image in Fig. 2(b) (ii), the dip-coated Ag NWs were entirely covered on the CC, forming a conductive nanonetwork. The diameter of each Ag NW was found to be $\sim 100 \pm 40 \text{ nm}$, as displayed in Fig. 2(b) (iii–iv). The as-fenced Ag NWs on CC not only serve as a conducting scaffold to enhance the mass loading of NC LDH NSs but also improve an effective electron transportation from the electroactive material to the current collector. After the growth of electroactive material on Ag@CC (Fig. 2(c) (i)), the color of the substrate changed into a light green, indicative of the successful deposition of NC LDH. The resulted NC LDH NSs@Ag@CC substrate possessed high flexibility as well (inset of Fig. 2(c) (i)). As shown in the FE-SEM image of Fig. 2(c) (ii), it is evident that the conductive network of Ag@CC fiber was densely wrapped with the electrochemically grown NC LDH material. From the magnified FE-SEM image in Fig. 2(c) (iii), the NC LDH is not only decorated on Ag NWs network but also on the interstitial regions of CC (indicated by white circles). Further increasing the magnification (Fig. 2(c) (iv)), the hierarchical

growth of NC LDH with interconnected NSs was clearly shown. Such hierarchical NS assembly on Ag@CC is more favorable for the rapid diffusion of electrolyte ions and enhances the electrochemical redox activity for improved energy storage properties [56]. Meanwhile, the FE-SEM images of the NC LDH NSs deposited on the bare CC (without Ag NWs) using the same ECD conditions are presented in Fig. S1 (a–b).

The morphologies of the as-prepared nanocomposite were further characterized by the FE-TEM analysis. To prepare the sample for TEM analysis, the CC substrate was scratched off with a scalpel and sonicated in ethanol solvent. The resulting solution was dropped on a copper grid and inserted into a TEM chamber. From the FE-TEM image in Fig. 3(a), it is clear that the Ag NWs were completely covered with the electrochemically grown NC LDH NSs. The as-grown NC LDH NSs on Ag NWs showed a hierarchical and interconnected NS assembly with a height of $\sim 70\text{--}90 \text{ nm}$ (from the outer surface of Ag NWs), as shown in Fig. 3(b). Moreover, the hierarchical NC LDH NSs showed an ultrathin thickness ($< 10 \text{ nm}$) (Fig. 3(c)). The HR-TEM image in Fig. 3(d) exhibited the lattice spacing width of 0.23 nm corresponding to the (200) plane of NC LDH NSs and it is in good agreement with the selected area electron diffraction (SAED) pattern (Fig. S2) [30]. The phase structure of the prepared samples was characterized using the XRD analysis and the corresponding patterns are demonstrated in Fig. S3. The broad peaks in the XRD patterns are related to the CC and the intense peaks are due to the deposited Ag NWs, respectively [57,58]. Meanwhile, the weak diffraction peaks observed at the 2θ values of 11.6° , 33.5° , 38.9° , 40.9° and 48.4° correspond to the α -nickel hydroxide hydrate phase (JCPDS # 22-0444) and the peaks located at the 2θ values of 19.6° , 32.4° and 51.3° are due to the presence of β -cobalt hydroxide (JCPDS # 30-0443) in the prepared materials. Moreover, some small peaks located at 24.5° , 26.9° , 29.8° and 45.6° reveal the existence of nickel cobalt oxy hydroxide (JCPDS # 29-0491). The presence of these diffraction peaks in the prepared samples indicates that successful formation of mixed metal hydroxide phases [30,38,59].

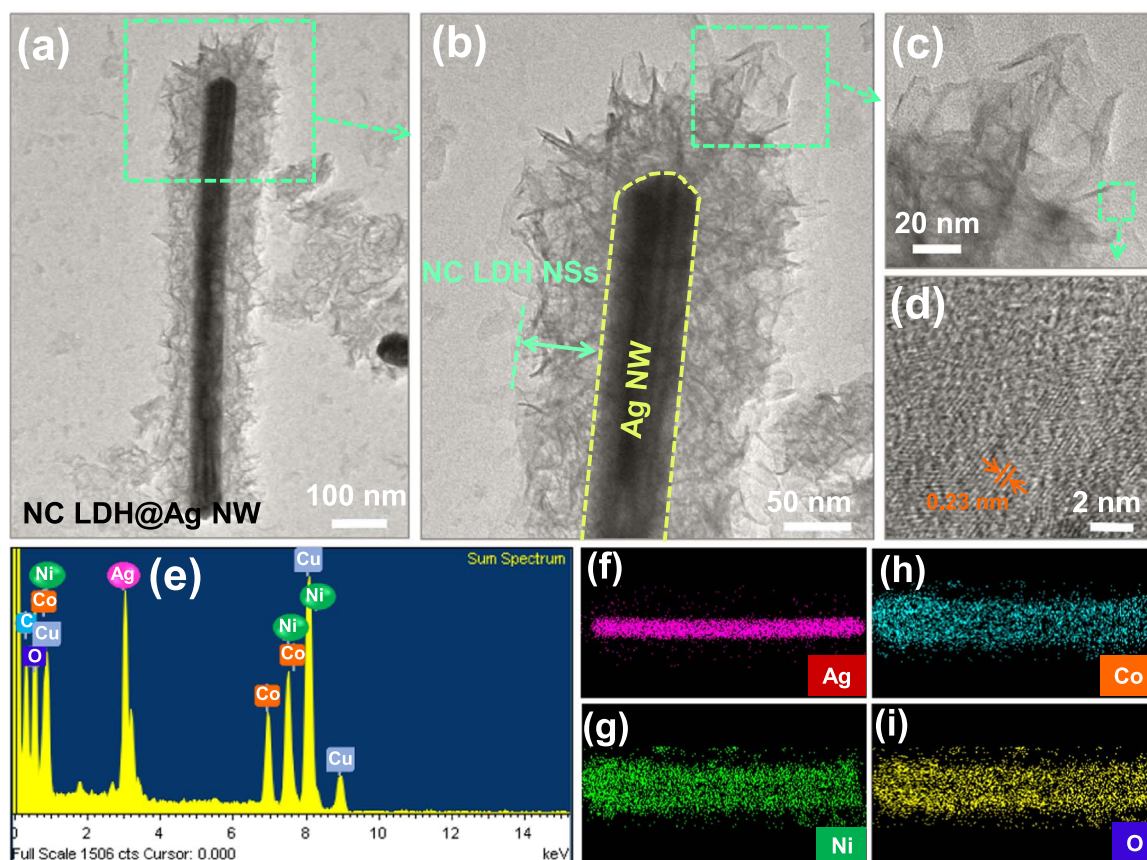


Fig. 3. (a–d) FE-TEM images, (e) EDX spectrum and (f–i) elemental mapping images of the NC LDH NSs coated Ag NW.

The elemental composition of the prepared sample was further analyzed by the EDX spectroscopy. In Fig. 3(e), the EDX spectrum displayed that the material composed of nickel, cobalt and oxygen peaks, indicating the successful deposition of hierarchical NC LDH NSs after ECD process. The other peaks representing the elements of Ag, Cu

and are attributed to the Ag NW fencing, CC substrate and TEM sample holder (copper grid), respectively. The corresponding mapping images of individual elements clearly revealed that the Ni and Co with O atoms were uniformly distributed on the Ag NWs without any impurities. Fig. S4 displays the specific surface area and pore size of the

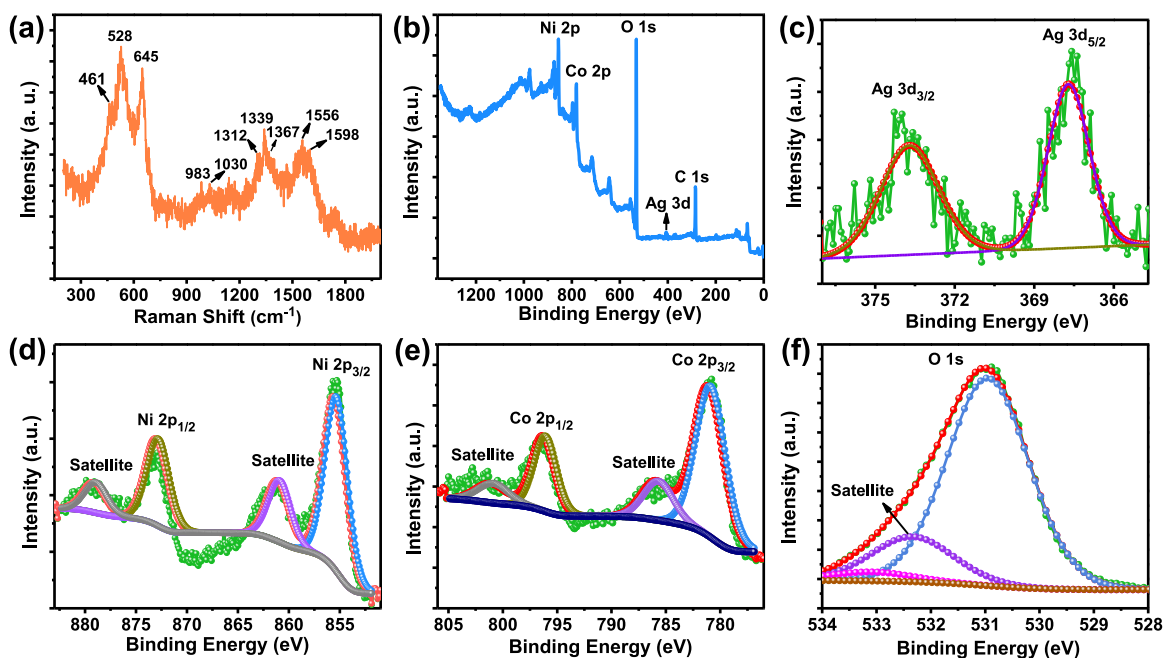
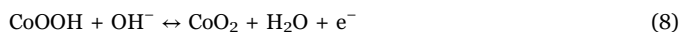
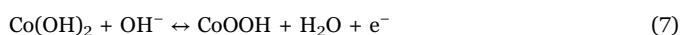
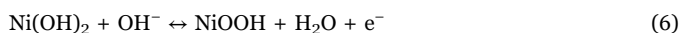


Fig. 4. (a) HR-Raman spectrum and (b–f) XPS analysis of the electrochemically grown NC LDH NSs@Ag@CC.

NC LDH NSs@Ag@CC. These values were calculated to be $30.1 \text{ m}^2 \text{ g}^{-1}$ and 1.96 nm , respectively, from the N_2 adsorption-desorption isotherm. This kind of hierarchical NC LDH NSs on Ag@CC with high surface area and pore distribution is believed to be helpful for obtaining high specific capacitance.

Fig. 4 shows the HR-Raman spectrum and XPS analysis of the prepared NC LDH NSs@Ag@CC. As shown in Fig. 4(a), the main peaks in the low Raman shift region (i.e., at 461 , 528 and 645 cm^{-1}) indicate the presence of E_g , F_2g and A_1g vibration modes of NC LDH [60]. The peaks observed at ~ 1339 and 1556 cm^{-1} are ascribed to the D and G bands of CC substrate, respectively [61]. Besides, the metallic silver is confirmed by the other peaks which are observed at 1312 and 1367 cm^{-1} in the HR-Raman spectrum [62]. The chemical compositions and surface binding energies of the prepared nanocomposite were identified by the XPS analysis. As can be seen in Fig. 4(b), the full survey scan spectrum of the NC LDH NSs@Ag@CC was predominantly composed of Ni 2p, Co 2p, O 1s, Ag 3d and C 1s peaks, confirming the existence of Ni, Co, O, Ag and C elements in the prepared sample. The two corresponding peaks of Ag $3d_{3/2}$ and Ag $3d_{5/2}$ were observed at 372.7 and 366.6 eV , respectively, as shown in Fig. 4(c), which indicates the existence of Ag metal in the sample [63]. The high-resolution XPS scans of both Ni 2p and Co 2p are shown in Fig. 4(d) and (e). The spin-orbit splitting into $2p_{1/2}$ and $2p_{3/2}$ with satellite peaks was identically demonstrated by these two (Ni 2p and Co 2p) spectral curves [30]. As shown in Fig. 4(a), the obtained two peaks of Ni $2p_{1/2}$ at 872.8 eV and Ni $2p_{3/2}$ at 855.3 eV correspond to the Ni^{2+} in nickel hydroxide phase. In the same manner, the Co $2p_{1/2}$ and Co $2p_{3/2}$ peaks at 796 and 780.9 eV in the Co 2p spectrum are due to the Co^{2+} valence states of cobalt hydroxide in the prepared material [64]. The O 1s core-level spectrum, as illustrated in Fig. 4(f), is composed of two peaks, i.e., one main peak at about 530.9 and other small peak at 532 eV , which are concerned to hydroxide ($-\text{OH}$) group [37]. Altogether, the XPS analysis obviously confirmed that the as-grown NC LDHs on Ag@CC was highly pure without any impurities.

The electrochemical properties of the binder-free NC LDH NSs@Ag@CC were studied by means of CV, GCD and EIS tests in 1 M KOH solution using the three-electrode system. As shown in Fig. 5(a), at a scan rate of 5 mV s^{-1} , both the electrodes including NC LDH NSs@CC and NC LDH NSs@Ag@CC exhibited a pair of redox peaks compared to the Ag@CC electrode in the potential range of -0.1 – 0.55 V . These redox peaks are ascribed to the faradaic electrochemistry behavior of NC LDH NSs [65,66]. The corresponding reversible faradaic reactions of NC LDH during the electrochemical measurements were given below [15]:



From the CV curves, it can be clearly compared that the NC LDH NSs@Ag@CC electrode exhibited much higher redox peak intensities and higher current values in CV curves than the NC LDH NSs@CC electrode. This means that the higher capacitance values could be expected for NC LDH NSs@Ag@CC electrode. Fig. 5(b) demonstrates the GCD curves of the Ag@CC, NC LDH NSs@CC and NC LDH NSs@Ag@CC electrodes at the constant current density of 1 mA cm^{-2} . It can be seen that in GCD curves, the redox behavior of Ag@CC electrode is negligible, indicating that the contribution of Ag@CC to the capacitance is small. Meanwhile, the NC LDH NSs@CC and NC LDH NSs@Ag@CC electrodes showed battery-type charge-discharge plots, signifying the faradaic redox reactions of NC LDH. Moreover, the NC LDH NSs@Ag@CC electrode exhibited longer charge and discharge times than the NC LDH NSs@CC electrode, confirming the higher energy storage properties. The longer charge-discharge time of the NC LDH

NSs@Ag@CC electrode is mainly attributed to the conductive framework of Ag NWs on CC electrode. Fig. 5(c) illustrates the Nyquist plots of the both samples. The EIS analyses were executed under an open circuit AC potential of 5 mV in the frequency range from 100 kHz to 0.01 Hz in 1 M KOH aqueous electrolyte. Both the spectral curves comprised of semicircle in the high-middle frequency region and sloped line (Warburg curve) in the low frequency region. The equivalent series resistance (R_s) of an electrode can be calculated from the intercept of semicircle with the real axis in the high-frequency region which is attributed to the internal resistance of an electrode material, electrolyte resistance and contact resistance between the active material and substrate, whereas the charge transfer resistance (R_{ct}) was obtained from the diameter of the semicircle [67]. Specifically, the NC LDH NSs@Ag@CC electrode exhibited the lower R_{ct} (1.78Ω) than that of the NC LDH NSs@CC electrode (R_{ct} of 2.24Ω), respectively, which further confirms the excellent conductivity of the NC LDH NSs@Ag@CC. Based on the above results, the NC LDH NSs@Ag@CC electrode was utilized to further investigate its higher electrochemical properties. The typical CV curves of the NC LDH NSs@Ag@CC electrode with various scan rates from 2 to 20 mV s^{-1} are shown in Fig. 5(d). The CV curves of an active material revealed a pair of well-defined redox peaks within the potential range, which clearly specifies the battery-type electrochemical behavior of active material. Meanwhile, the changes of anodic and cathodic peaks towards more positive and negative directions can be observed with increasing the scan rate from 2 to 20 mV s^{-1} . This is due to the increased internal resistance of the prepared electrode under higher scan rates. Furthermore, the GCD tests of the NC LDH NSs@Ag@CC electrode were conducted at different current densities ranging from 1 to 15 mA cm^{-2} with the potential window of -0.1 – 0.45 V in the 1 M KOH aqueous electrolyte (Fig. 5(e)). The symmetric nature and non-linear shaped GCD curves are in well agreement with the CV curves and fast faradaic reversibility of the prepared material. The GCD curves of the NC LDH NSs@CC electrode measured under various current densities are also shown in Fig. S5. The calculated areal capacitance values from the GCD curves of the both materials are plotted in Fig. 5(f). At a current density of 1 mA cm^{-2} , the areal capacitance of the NC LDH NSs@Ag@CC electrode was $1133.3 \text{ mF cm}^{-2}$ which is about 2.3 times higher than the NC LDH NSs@CC electrode (486.2 mF cm^{-2}). The corresponding areal capacitance values at various current densities for the NC LDH NSs@Ag@CC electrode were 1133.3 , 1117.7 , 1095.3 , 1067.5 , 1057.1 , 1035.6 and $1032.2 \text{ mF cm}^{-2}$ at the current densities of 1 , 2 , 3 , 5 , 7 , 10 and 15 mA cm^{-2} , respectively. And the observed areal capacitance values for the NC LDH NSs@CC electrode were 486.2 , 460.2 , 450.2 , 436.8 , 425.1 , 416.3 and 409.6 mF cm^{-2} under the same current densities, respectively. The lower electrochemical properties of the NC LDH NSs@CC are mainly due to the small mass loading, poor conductivity and high hydrophobic nature of CC substrate. The corresponding surface compatibility properties of the electrodes are shown in Fig. S6. Moreover, the obtained energy storage performance of NC LDH NSs@Ag@CC electrode is competitive with the previously reported similar kind of bimetallic LDH nanomaterials, as shown in Table S2. Cycling stability is a crucial factor for any electrode materials. The cycling performance of the NC LDH NSs@Ag@CC electrode was tested in 1 M KOH electrolyte at the constant discharge current density of 4 mA cm^{-2} for 2000 cycles in a three-electrode configuration, as shown in Fig. 5(g). The NC LDH NSs@Ag@CC electrode showed high areal capacitance during the initial cycles, but the capacitance decreased slowly during the extended cycles and still retained about 80.47% of its original capacitance after 2000 successive charge-discharge cycles, indicating a good stability. Even after a long term cycling test, as shown in the FE-SEM images (inset of Fig. 5(g)), the NC LDH NSs were still intact on Ag NWs without being peeled out from the CC fibers. Moreover, these NC LDH NSs well maintained its hierarchical structure, which exhibits the good stability of an active material. To further demonstrate the morphology and structure of the NC LDH

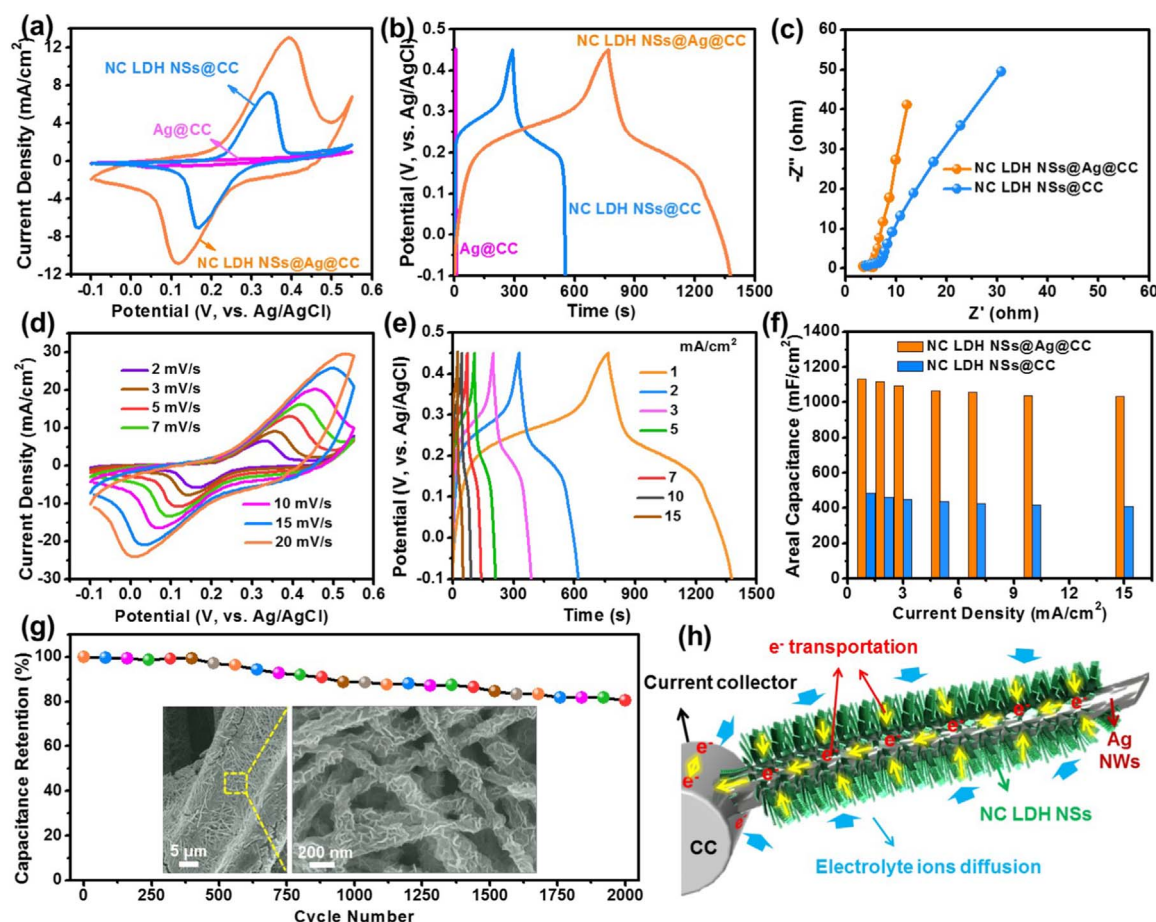


Fig. 5. (a–c) Comparative CV, GCD and EIS analyses of the Ag@CC, NC LDH NSs@Ag@CC and NC LDH NSs@CC electrodes in 1 M KOH electrolyte solution. (d–e) CV and GCD curves of the NC LDH NSs@Ag@CC electrode at various scan rates and current densities. (f) Calculated areal capacitance values as a function of applied current density for the NC LDH NSs@Ag@CC and NC LDH NSs@CC electrodes. (g) Cycling stability of the NC LDH NSs@Ag@CC electrode at a current density of 4 mA cm⁻². (h) Schematic diagram showing the beneficial usage of the NC LDH NSs@Ag@CC electrode for high-performance energy storage properties.

NSs@Ag@CC, we performed the XRD and EDX analyses of the electrode after cycling test (Fig. S7), indicating that there is no noticeable change in the phase and purity of the material. The desirable advantages of the NC LDH NSs@Ag@CC electrode during the electrochemical measurement are schematically shown in Fig. 5(h). The hierarchical NC LDH NSs with large surface area and interconnected network can allow for the diffusion of electrolyte ions effectively into the NSs and the rapid movement of electrons from NSs to CC substrate can be achieved by the intermediate Ag NWs. Hence, the Ag NWs and CC offer a fast transportation of released electrons during the charging and discharging processes, which leads to the increased areal capacitance, high rate performance and long cycling stability.

For practical applications, the fabrication of ASC with high energy storage performance is desirable. Fig. 6(a) shows the schematic diagram of the fabricated ASC using NC LDH NSs@Ag@CC as a battery-type electrode, commercial AC coated CC as an electric double layer capacitive electrode and cellulose based filter paper as a separator, respectively, in 1 M KOH electrolyte. Initially, both the electrodes were soaked in 1 M KOH along with filter paper and firmly packed by the help of a Kapton tape. Commercially available AC is a good material due to its large negative potential window (–1.0 V) and long cycling lifetime. The electrochemical properties (CV and GCD) of the AC are displayed in Fig. S8. Before assembling the device, the mass on both electrodes was optimized based on the following equation [68]:

$$\frac{m^+}{m^-} = \frac{C^- \times \Delta V^-}{C^+ \times \Delta V^+} \quad (9)$$

Here, m^+ , C^- and ΔV^- are the mass, areal capacitance and potential window of the AC coated CC, and m^+ , C^+ and ΔV^+ are the mass, areal capacitance and potential window of the NC LDH NSs@Ag@CC electrode, respectively. The mass loading of positive (NC LDH NSs on Ag@CC) and negative (AC@CC) electrodes was optimized to be ~ 0.7 mg/cm² and ~ 0.9 mg/cm², respectively. To estimate the best operating potential of the as-prepared device, the CV curves of the NC LDH NSs@Ag@CC device were studied at different potentials ranging from 0.5 to 1.6 V at 30 mV s⁻¹, as shown in Fig. 6(b). This demonstrates that the potential window of the as-constructed device was gradually extended to 1.6 V without altering the shape of curves. In Fig. 6(c), it can be clearly observed that the GCD curves with relatively symmetrical triangular shapes, which are collected at different potentials from 0.5 to 1.6 V at the current density of 2 mA cm⁻², are indicative of the good capacitive behavior of the device. From the plateaus of CV and GCD curves at different potentials, it is concluded that the optimum potential window of the NC LDH NSs@Ag@CC device was 1.6 V. Therefore, an electrochemical performance of the assembled device was further tested with a potential of 0–1.6 V. Fig. 6(d) shows the CV curves of the ASC measured at various scan rates from 5 to 100 mV s⁻¹. In contrast to the redox peaks, the obtained CV curves displayed the quasi-rectangular shapes, which are due to the presence of both electric double layer and battery-type materials. To further examine the performance of the device, the GCD measurements were also carried out in the potential range of 0–1.6 V (Fig. 6(e)). Except at low current density (1 mA cm²), the obtained symmetrical GCD curves elucidated the well balancing mass between the both

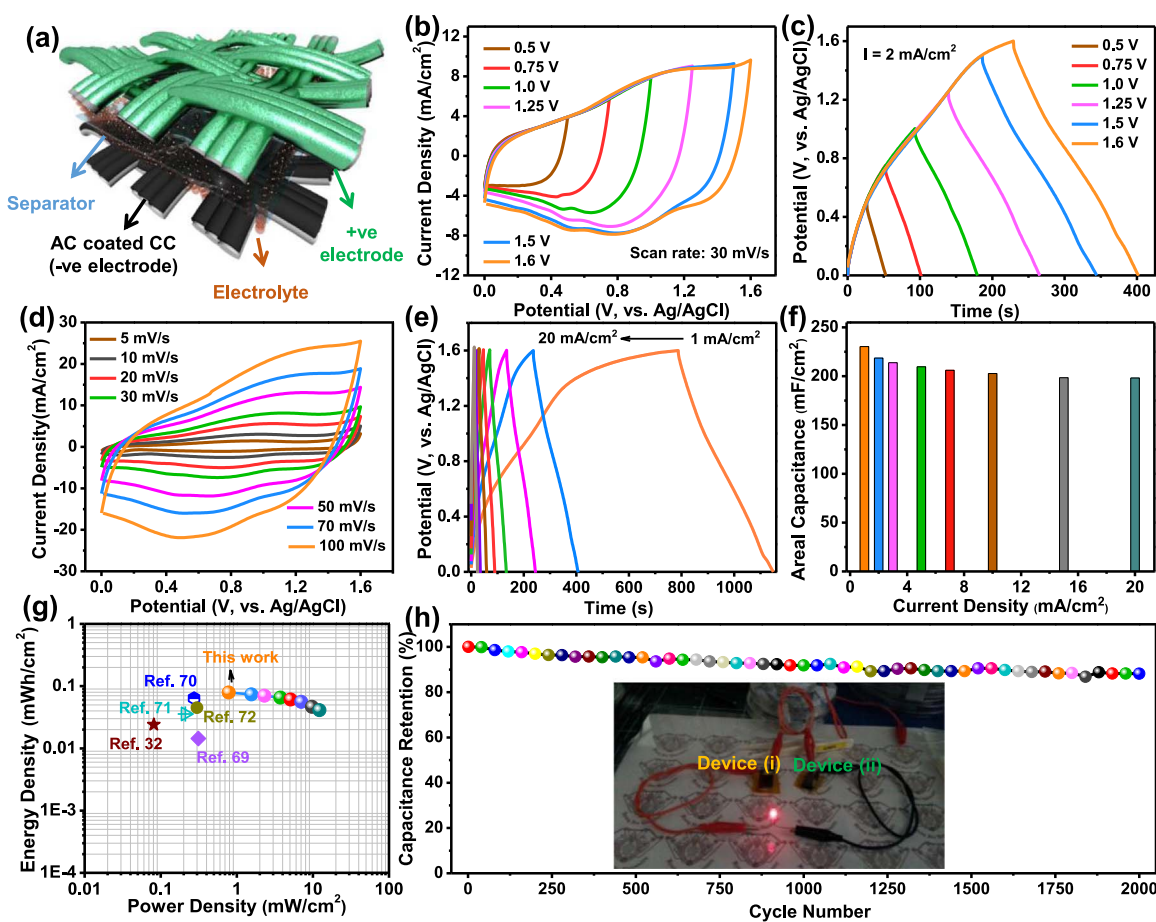


Fig. 6. (a) Schematic illustration of the fabricated ASC with NC LDH NSs@Ag@CC and AC coated CC electrodes. (b–c) CV and GCD curves of the ASC operated at different potentials. (d–e) CV and GCD curves of the ASC measured at different scan rates and current densities under the fixed potential window of 0–1.6 V. (f) Calculated areal capacitance values as a function of applied current density for the ASC device. (g–h) Ragone plot and cycling stability of the ASC device. The inset of (h) shows the photographic image of two serially connected ASC devices glowing a red-colored LED.

electrodes in the device. As a function of current density, the areal capacitance values (Fig. 6(f)) of the ASC device calculated from the GCD curves were about 230.2, 218.5, 213.8, 209.6, 206, 202.7, 198.4 and 198.1 mF cm^{-2} at 1, 2, 3, 5, 7, 10, 15 and 20 mA cm^{-2} , respectively. Besides, the energy and power densities are the two important factors for evaluating the performance of the SCs. The Ragone plot in Fig. 6(g) denotes the areal energy and power densities of the NC LDH NSs on Ag@CC//AC device, which are calculated from the Eqs. (2) and (3). At a current density of 1 mA cm^{-2} , the device exhibited a maximum areal energy density of 78.8 $\mu\text{Wh cm}^{-2}$ with a power density of 785 $\mu\text{W cm}^{-2}$. As the current density increased to 20 mA cm^{-2} , the energy density of the ASC was still obtained as 40 $\mu\text{Wh cm}^{-2}$ with a high power density of 12.1 mW cm^{-2} . Satisfactorily high energy density was delivered even at high power density, which demonstrated the excellent performance of the assembled device. Moreover, these values are superior than for recently reported asymmetric SCs: Ni-Co LDH//ketjen black (14.4 $\mu\text{Wh cm}^{-2}$ at 312.5 $\mu\text{W cm}^{-2}$) [69], Mg-Al LDH//rGO (24.07 $\mu\text{Wh cm}^{-2}$ at 81.2 $\mu\text{W cm}^{-2}$) [32], Ag_3O_4 grafted NiO//rGO (64.03 $\mu\text{Wh cm}^{-2}$ at 275 $\mu\text{W cm}^{-2}$) [70], $\text{Co}_{0.85}\text{Se}/\text{AC}$ (35.65 $\mu\text{Wh cm}^{-2}$ at 210 $\mu\text{W cm}^{-2}$) [71] and $\text{Ni}_{0.76}\text{Co}_{0.24}\text{LDH}/\text{AC}$ (45.01 $\mu\text{Wh cm}^{-2}$ at 300 $\mu\text{W cm}^{-2}$) [72]. The long-term cycling stability of the ASC device was also carried out at a current density of 5 mA cm^{-2} up to 2000 cycles, as shown in Fig. 6(h). The cycling test of the as-assembled device was extended steadily with slight variations. After 2000 cycles, the device exhibited a capacitance retention of 88.1%, specifying the high stability of the device. Considering the high energy and power density, the serially connected

two ASC devices lighted up the commercial red light-emitting diode (LED) for 160 s, as displayed in the inset of Fig. 6(h). Finally, the flexibility of the ASC device was performed by bending the test device under various deformations. Fig. S9 displays the CV curves (scan rate of 30 mV s^{-1}) and calculated areal capacitance values of the corresponding device under flat and flexed conditions. It can be seen that there is slight increment in the capacitance when flexed and the CV shapes were almost unchanged in all the conditions. In the case of flexed conditions, the distance between the both electrodes decreases, which intends to improve the electrochemical conductivity and ion transportation. These results indicate the excellent flexibility and electrochemical stability of the device under flexed conditions. By virtue of superior areal capacitance, high energy/power densities and long-term cycling stability, the assembled ASC (NC LDH NSs@Ag@CC//AC) device is expected to be utilized for flexible and high-performance energy related applications.

4. Conclusion

In summary, we have successfully deposited the NC LDH NSs@Ag@CC by a single-step ECD process. Here, the Ag NWs serve as a conducting scaffold to improve the conductive nature and hydrophilicity nature of the CC, which initiated the high mass loading of NC LDH NSs. The electrochemical properties of the NC LDH NSs@Ag@CC electrode exhibited the superior areal capacitance of 1133.3 mF cm^{-2} at 1 mA cm^{-2} than that of the NC LDH NSs@CC electrode (486.2 mF cm^{-2}) at a current density of 1 mA cm^{-2} . Furthermore, an ASC was

assembled with NC LDH NSs@Ag@CC and AC coated CC in 1 M KOH aqueous electrolyte. The electrochemical properties suggested that the ASC delivered a high energy density of $78.8 \mu\text{Wh cm}^{-2}$ and the power density of 12.1 mW cm^{-2} with the high potential window of 1.6 V. Furthermore, the device showed a good cycling stability with a capacitance retention of 88.1% after 2000 cycles. Such impressive properties of the NC LDH NSs@Ag@CC electrode and assembled ASC could be expected to serve as a promising platform for the development of flexible energy storage devices.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at <http://dx.doi.org/10.1016/j.nanoen.2017.04.019>.

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